Junhuai Xu

Ph.D. Candidate, Experimental Nuclear Physics
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Curriculum Vitae
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Profile

Ph.D. candidate in experimental nuclear physics developing reconstruction and inference methods for complex detector data. My research connects detector calibration, response simulation, inverse problems, statistical inference, and physics-informed machine learning across nuclear, heavy-ion, and neutrino experiments. I have published five first-author peer-reviewed articles.

Research Experience

Nucleon Short-Range Correlations Oct. 2023 – Present
CSHINE Experiment, RIBLL-1 (HIRFL), Lanzhou, China

- Within the IBUU-MDI framework, constrained the SRC-induced high-momentum-tail fraction in ^{124}Sn within the IBUU-MDI framework to $R_{\mathrm{HMT}} = (20 \pm 3)\%$ using precision hard-γ spectra and detector-filtered transport calculations.
- Developed calibration and detector-response models for the CsI(Tl) γ-ray hodoscope used in bremsstrahlung measurements from low-energy heavy-ion collisions.
- Implemented Richardson–Lucy spectrum unfolding and quantified reconstruction/systematic uncertainties through pseudo-experiments, closure tests, and consistency checks.

Correlation Function Imaging Feb. 2024 – Present
Department of Physics, Tsinghua University

- Developed a Richardson–Lucy framework that directly images femtoscopic source functions without imposing a Gaussian shape and simultaneously constrains strong-interaction parameters.
- Reconstructed non-Gaussian source functions from published pp and $\bar{p}\bar{p}$ correlations in Au+Au collisions at $\sqrt{s_{NN}} = 200$ GeV; the two source functions agree within uncertainty.
- Contributed to a published three-dimensional extension for pion-source imaging and neutron-skin sensitivity studies in heavy-ion collisions.

Deep-Sea Neutrino Detection Jun. 2024 – Present
TRIDENT Collaboration; first-author manuscript prepared for JINST submission

- Built an atmospheric-neutrino simulation-to-reconstruction chain combining HONDA fluxes, FLUKA interactions, and Geant4 detector transport for a compact underwater detector concept motivated by TRIDENT.
- Developed a PyTorch Geometric graph-neural-network workflow using geometry-aware graph construction and Edge Convolution to reconstruct event topology, direction, and energy from sparse detector responses.
- Connected reconstructed outputs to oscillation-parameter inference over 64,273 selected events and recovered the input parameters within uncertainty through a physics-level closure test.

Experimental Experience

- **CSHINE Experiment** (Lanzhou, China, 2024): participated in beam-time data taking and led calibration, event reconstruction, and spectrum extraction for the high-energy γ-ray detector array.
- **SLEGS Beam Test at SSRF** (Shanghai, China, Jan. 2025): designed and led a dedicated CsI(Tl) response experiment from detector testing and setup through beam measurement, Geant4 calibration, and analysis.
- **SπRIT Experiment** (RIKEN, Japan, 2024): supported detector operation and electronics control during SAMURAI beam time, contributing to stable data taking and run coordination.

Technical Skills

Physics analysis	Detector calibration, detector-response simulation, event reconstruction, spectrum unfolding, systematic uncertainty evaluation.
Inference	Inverse problems, Richardson–Lucy deconvolution/imaging, likelihood inference, particle filtering, unscented Kalman filtering, pseudo-experiments, closure tests.
Machine learning	PyTorch, PyTorch Geometric, graph neural networks, Edge Convolution, LoRA/PEFT, physics-informed reconstruction.
Computing	Python, C/C++, ROOT, Geant4, FLUKA-based workflows, Linux, Git, HTCondor, HPC/GPU workflows, LaTeX.
Experiment	CAEN-based electronics operation, data acquisition, detector testing, beam-time operation, and run coordination.

Education

Tsinghua University Aug. 2022 – Jun. 2027 (Expected)
Ph.D. Candidate in Physics; specialization in Experimental Nuclear Physics
Research areas: short-range correlations, heavy-ion collisions, femtoscopy, neutrino simulation, and machine-learning-based reconstruction.

South China Normal University Aug. 2018 – Jun. 2022
B.S. in Optoelectronic Information Science and Engineering
GPA: 4.06/5.00; Rank: 1/137. Undergraduate thesis in DIS/SIDIS phenomenology: *A Study of the Nuclear Medium Effect of Hadron Fragmentation Function*.

Selected Publications

Five first-author peer-reviewed articles; complete publication record available on Google Scholar.

- **Xu, J.** *et al.*, "Experimental study of bremsstrahlung γ-ray emission and short-range correlations in ^{124}Sn+^{124}Sn collisions at 25 MeV/nucleon," ***Phys. Rev. C*** **113**, 044613 (2026).
- **Xu, J.** *et al.*, "Precise measurement of short-range correlations in nuclei from bremsstrahlung gamma-ray emission in low-energy heavy-ion collisions," ***Phys. Rev. Res.*** **7**, 043174 (2025).
- **Xu, J.** *et al.*, "Linear response of CsI(Tl) crystal to energetic photons below 20 MeV," ***Nucl. Instrum. Meth. A*** **1080**, 170787 (2025).
- **Xu, J.** *et al.*, "Imaging Freeze-Out Sources and Extracting Strong Interaction Parameters in Relativistic Heavy-Ion Collisions," ***Chin. Phys. Lett.*** **42**, 031401 (2025).
- **Xu, J.** *et al.*, "Reconstruction of Bremsstrahlung γ-rays spectrum in heavy ion reactions with Richardson-Lucy algorithm," ***Phys. Lett. B*** **857**, 139009 (2024).

Selected Presentation

- **5th International Workshop on Quantitative Challenges in SRC and EMC-Effect Research** (LBNL, remote, Jun. 2026): "Experimental Study of Bremsstrahlung Gamma-Ray Emission and Short-Range Correlations in Low-Energy Heavy-Ion Collisions."

Mentoring and Teaching

- Mentored an undergraduate project on a plastic-scintillator cosmic-ray detector and an undergraduate thesis on GNN-based neutrino-event topology classification.
- Outstanding Teaching Assistant, Tsinghua University (2024).

Awards

- First-class Comprehensive Scholarship, Tsinghua University (2025).
- Second-class Comprehensive Scholarship, Tsinghua University (2024).
- Third Prize, National Undergraduate Mathematics Competition Final Round, Non-Math Major (2021).
- First Prize, China Undergraduate Mathematical Contest in Modeling (CUMCM, 2020).